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The Green Blueprint: Founder Clan Culture Imprints and Green Innovation in Chinese Family Firms

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Abstract: Drawing on upper-echelon, imprinting, and agency theories, this paper investigates the influence of founder CEOs' clan cultural values on family firms' green innovation performance and examines the moderating roles of family involvement and the intensity of government environmental regulation. Using data from 1678 Chinese public family firms over the period from 2012 to 2020, which include 9048 firm–year observations, our results show a significant positive effect of founder CEOs' clan cultural values on firms' green innovation practices. Additionally, this positive effect is weakened when the level of family involvement is high, due to conflict. Moreover, the positive influence is strengthened when government environmental regulation is more stringent.

Keywords: founder imprints; clan cultural values; green innovation; family involvement; government environmental regulation

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1. Introduction

In an era where environmental challenges are global in scale, the propensity of a firm to engage in green innovation is increasingly recognized as a strategic imperative. Green innovation, defined as technological advancement aimed at energy efficiency, eco-friendly design, and improved environmental practices [1], offers companies a chance to boost economic growth and sustainability simultaneously [2]. As the most widespread form of business organization around the world [3], family firms provide an interesting context in which to study environmentally related issues, such as green innovation, due to their unique characteristics, which include prioritizing family goals and long-term orientations [4]. Considering the substantial impact of family firms on global GDP, coupled with their distinctive characteristics, it is imperative to scrutinize the precursors of their green innovations.

Scholars in management have aimed to explore the motivations behind firms' heterogeneous performance in green innovation [5] and have identified external forces of green innovation through different theoretical lenses, including stakeholder greening demands [6], environmental regulations [7], and institutional pressure [8]. However, comparatively less attention has been paid to firms' internal drivers, especially the effects of founder CEOs' personal motivations [9]. Drawing on upper-echelon and imprinting theories, a body of research has demonstrated how the personal traits of founder CEOs can have a lasting

impact on their organizations, influencing their structure, procedures, and culture and thus directing the company's path [10–12]. Although prior studies have explored the effects of various CEO characteristics, such as unique life experiences [13,14] and personal ideologies [15], the literature still lacks an in-depth examination of the influence of CEOs' cultural values on green innovation. Given the significant explanatory power of cultural factors on CEOs' perceptions and strategic decisions, this research gap is striking.

In this paper, we focus on the effects of founder CEOs' clan cultural values, known as 'Zong Zu Wen Hua', which prioritize the continuity of family lineage, reputation, communal well-being, and long-term orientation and are substantial and prevalent sub-national cultural values in Asian clan cultures [16,17]. Despite the social and economic changes brought about by urbanization and industrialization, clan culture still exerts significant influences on people's and organizations' behaviors [17]. For example, ALPHA Entertainment, a well-known and promising animation and cultural enterprise in China, was founded by the Cai Dongqing family in Shantou, Guangdong Province, a region where family ties and clan culture are deeply valued. The principles embedded in clan culture emphasize harmony within both the family and the broader community, which likely influenced Cai' s approach to developing ALPHA Entertainment into a company that seeks to balance financial success with cultural responsibility. Cai's leadership in an industry that combines cultural heritage with modern innovation can be viewed as a reflection of the family-oriented, long-term vision typical of the region's clan culture. The company's success in blending traditional Chinese cultural elements with modern animation technologies is a direct reflection of the founder's long-term, family-oriented thinking, demonstrating how clan values can shape a firm's strategic orientations. Members of an organization in a clan culture typically view themselves as integral components of a large, unified family. Within the cultural framework of clan organizations, leaders frequently adopt roles that resemble those of parental figures, a dynamic that is especially crucial in family business settings. Family-owned businesses tend to embody a clan culture more so than non-family counterparts, with their corporate culture being deeply shaped by the values and traditions of the family and founders [18].

Drawing on upper-echelon and imprinting theory, we propose that founder CEOs' clan culture imprints are positively associated with green innovations because they emphasize collective well-being, family succession, and long-term orientation [16,17], thus driving the firm towards more sustainable practice and green innovation. In line with the executive job demands literature [19], we also document that the effect of a founder's clan cultural imprints is not uniform or systematic, being constrained by both family dynamics and the external political environment in which they are embedded. On one hand, CEOs are not only influenced by the prevailing national culture, but their decision-making processes are also made more complex by their social interactions with family members [20]. Family businesses, in particular, operate in a context where the personal traits and values of the founder—especially those related to clan cultural values—are intertwined with the family's collective goals, values, and interpersonal dynamics [21,22]. Based on socioemotional wealth and behavioral agency theory [23,24], we examine the moderating role of family involvement. On the other hand, government environmental regulation is identified as an important contextual factor for examining the relationship between executives and environmental practices, based on institutional logic [25].

Our study enriches the body of literature in multiple aspects. First, we elucidate the effect of founder CEOs' clan cultural values on firms' green innovation, highlighting the significance of founders' psychological traits and their impact on firm-level decisions and outcomes, which is a central tenet of upper-echelons theory [26]. Beyond external socioeconomic drivers, we argue for the critical role of clan cultural values at the upper-

echelon level in understanding the diverse environmental practices of firms, particularly in the Asian context. While much of the upper-echelon literature has traditionally focused on CEOs' demographic characteristics (e.g., age, gender, tenure) [27,28], personal traits (e.g., hubris, narcissism) [29], and professional experiences (e.g., international exposure) [14] as predictors of corporate green innovation, our study significantly extends this body of work by incorporating cultural values, with a particular emphasis on the clan-oriented values of CEOs. This theoretical expansion not only enriches the green innovation literature but also offers a more nuanced understanding of how deeply ingrained cultural factors influence the strategic decisions of top executives, shaping firms' heterogeneous environmental practices. Second, we identify the moderating roles of family involvement and government environmental regulation in the relationship between founders' value imprints and green innovation, which contributes empirical evidence to the understanding of the dynamic process of family firms' imprints changing [11]. We argue that founders' clan culture imprints may evolve due to family involvement and changes in government policy, adding to the founder imprints literature [10,30]. By discussing the interaction between founder CEOs' clan cultural values and family involvement, our paper contributes to the literature on family businesses by deepening the understanding of agency problems arising from heterogeneous family-centered goals [31]. Specifically, we argue that high levels of family involvement can give rise to multiple, often conflicting, family-centered objectives, which in turn dilute the unified socioemotional wealth (SEW) goals typically associated with high-risk, innovative decisions in family firms. This contribution expands the literature on the heterogeneity of family firms [31].

2. Literature Review and Hypotheses Development

2.1. Upper-Echelon and Imprinting Theories

Upper-echelon theory suggests that the variability in organizational behaviors and outcomes is largely due to the diversity among top executives, notably CEOs [26]. Their experiences, values, and personalities play a crucial role in shaping strategic choices and firm performance [32]. Imprinting theory adds that an organization's early conditions have a lasting effect on its trajectory [12], with Marquis and Tilcsik (2013) [10] highlighting the enduring imprint of a founder's values on organizational behavior and success. By merging upper-echelon and imprinting theories, we suggest that CEOs' cultural values critically influence firms' strategic decisions including in green innovation.

2.2. Clan Culture Imprints on Founder CEOs

The current literature on culture underscores the pivotal role of national culture in influencing a firm's behavior toward green innovation, often using Hofstede's model as a theoretical basis [33]. For example, Marquis and Battilana (2009) [34] suggest that collectivist values can promote collaborative approaches to green innovation. However, most studies have focused on Western contexts, lacking in-depth analysis of Asian settings and the influence of sub-national over broad national cultures. While some studies have examined the impact of cultural values on firm strategies within Asian contexts, they have predominantly focused on Confucian values or 'guanxi' to elucidate cultural influences [35]; the effects of clan culture are largely ignored.

The influence of clan culture, often overlooked, plays a critical and enduring role in family business behavior [17]. The clan system, a key social institution in ancient Asian societies, especially China, continues to significantly impact the patterns of industrial specialization across Chinese prefectures today [36]. A clan is a community organized around familial ties, primarily through paternal lineage, purportedly descending from a common male ancestor [37]. This cultural framework emphasizes familial bonds, collective

ancestor veneration, the construction of ancestral halls, the adoption of common surnames to signify lineage, and the detailed recording of family histories [38]. Clan culture influences various facets of Chinese society and economy, including entrepreneurship [39], business performance [40], and strategies [17].

Founder CEOs who adhere to clan cultural values tend to preserve their family's hereditary line by its constituents and have strong desire to accumulate collective wealth and ensure its intergenerational transfer to benefit their descendants. Therefore, founders deeply rooted in clan culture are more likely to have long-term orientation and maintain family business succession [41]. The tendency of clan-oriented founders to maintain family business succession is further supported by the socioemotional wealth perspective, which suggests that family leaders derive identity, satisfaction, and emotional benefits from their control and influence over a firm [23].

2.3. Founder CEOs' Clan Cultural Values and Firm's Green Innovation

Building on upper-echelon and imprinting theory, we propose that founder CEOs' clan cultural values inform the cognitive framework of an organization and thus affect its green innovation initiatives [6]. As discussed above, as clan-oriented founder CEOs are invested in the longevity of their lineage and the perpetuation of their family's wealth, they are naturally inclined to consider the long-term impact of their actions on their firm's viability. Founder CEOs steeped in clan values are more likely to invest in long-term strategies that prioritize the endurance and intergenerational success of their family businesses, rather than focusing on short-term gains [41]. As environmental degradation poses a risk to the future of any business that relies on natural resources and a stable climate, clan-oriented founder CEOs tend to treat green innovation as a promising long-term strategy to achieve both economic growth and environmental protection.

Additionally, clan culture's emphasis on collective identity and reputation [42] encourages founder CEOs to maintain a positive public image, which today is increasingly tied to environmental responsibility. Greif and Tabellini (2010, p. 136) [36] contend that cooperation within a clan is primarily upheld by moral obligations and reputational incentives, which deter dishonesty and free-riding behaviors. As consumers and society at large become more environmentally conscious, a firm's commitment to green innovation becomes a critical component of its reputation and competitive advantage [8]. Consequently, founder CEOs who are profoundly influenced by clan culture are more inclined to proactively pursue green innovation initiatives, thereby safeguarding and enhancing their family's prestige and the firm's standing in a competitive, eco-conscious landscape. Taking these together, we hypothesize the following:

Hypothesis 1. *A founder CEO's clan cultural values have a positive effect on a firm's green innovation practices.*

2.4. The Moderating Role of Family Involvement

Clan cultural values in the context of a family business often revolve around unity, loyalty, and the welfare of the family collective [22]. A founder CEO imbued with such values typically seeks to infuse the business with a sense of family goals and longevity. However, when the involvement of family members in management is extensive, the potential for conflict increases, and aligning family and business goals can become more challenging [43]. Family members, often acting as key stakeholders, can either reinforce or inhibit the CEO's decisions, depending on their own values, interests, and preferences [44]. This makes family involvement a crucial moderator of a CEO's cultural influences on firm behavior, as the need to balance family interests with organizational goals often results in decision-making that prioritizes family cohesion over strategic innovation, particularly

in high-stakes areas such as green innovation [43]. Miller and colleagues (2014) [45] demonstrated how the context of family firms influences the relationship between a CEO's characteristics and product innovation, highlighting that family goals are the essential factor guiding the behavior of family firms. Building on this framework, we suggest that the effect of a founder CEO's clan cultural values on green innovation may be moderated by the level of family involvement in management.

According to SEW theory, family firms prioritize non-financial goals, such as maintaining family control, preserving family identity and reputation, and ensuring family legacy, over purely financial outcomes [23,46]. Deep and entrenched family involvement heightens a family's commitment to preserving SEW. Family managers may fear the risks associated with green innovation investments, as these could jeopardize their control over the firm [47,48].

Behavioral agency theory expands on traditional agency theory by recognizing that family firms exhibit distinct behavioral traits that influence strategic decision-making, particularly in situations of conflict between personal interests and organizational goals [49]. Previous research has indicated that firms with greater family involvement encounter challenges in enhancing their performance due to conflicts of interest among different family factions, regardless of external ownership [50]. These challenges manifest as agency costs, which include higher levels of entrenchment [51] and lead to a decrease in a firm's value [52]. High family involvement in a business can create complex management and governance dynamics, as extended family members with different sub-family interests gain influence, potentially leading to varied objectives and priorities.

Agency costs in family firms arise due to conflicts of interest among family members who may have different objectives and levels of involvement in the firm [51]. For example, non-active family members may prioritize dividends over re-investment in innovation [53]. This can lead to underinvestment in green innovation due to the preference for short-term financial gains over long-term sustainability [25]. This situation poses a particular challenge to the implementation of a founder CEO's clan cultural values, which prioritize the collective interests of the larger family or clan. The conflict between collective clan values and individual self-interests may lead to a dilution of the effect of a founder's value, particularly in environmentally sustainable practices, which typically require long-term investment and a willingness to forego immediate returns [54]. High family involvement can lead to misalignment between individual family members' goals and the founder's vision, hindering green innovation. Increased family presence may boost rent-seeking behaviors, with family members prioritizing personal gains over investing in green innovation [50,55], diminishing the founder CEO's cultural values' impact on green innovation.

Therefore, we hypothesize as follows:

Hypothesis 2. *Family involvement weakens the positive influence of a founder CEO's clan cultural values on a firm's green innovation.*

2.5. The Moderating Role of Government Environmental Regulation

The intensity of government environmental regulations, which is defined as the stringency and scope of government-enacted policies and measures aimed at environmental conservation [56], can significantly influence the relationship between CEOs' values and green innovation by providing a formal framework that promotes sustainable practices. CEOs who value clan culture and endorse communal well-being may find such frameworks validating and supportive of their predispositions towards green innovation [57]. When governments enforce stringent environmental policies, it sends a clear message about the state's commitment to sustainable development, which can encourage clan-oriented CEOs to align their organizational strategies with these values, leading to an increased focus on green innovation to meet or exceed regulatory standards [58].

In addition, the intensity of government environmental regulations often dictates the resources available for firms' green innovation. For example, grants, tax incentives, and subsidies provided as part of these regulations can ease the financial burden associated with green innovation, particularly in research and development (R&D) [59]. This can be particularly appealing to CEOs with clan cultural values, as it aligns with their preference for cooperative efforts and mutual support, as well as their long-term orientation towards the community's sustainability [60], thus contributing to green innovation. Taking these together, government environmental regulation intensity serves as a critical stimulating engine for founder CEOs rooted in clan culture to invest in green innovation by providing a supportive framework, signaling the importance of sustainable corporate practice and offering valuable recourse. Therefore, we hypothesize as follows:

Hypothesis 3. *The intensity of government environmental regulation strengthens the positive influence of a founder CEO's clan cultural values on a firm's green innovation.*

3. Method

3.1. Data and Sample

Our hypotheses were tested using data from publicly trading family businesses in China, spanning from 2012 to 2020. This information was obtained from the China Stock Market and Accounting Research database (CSMAR), a leading provider of financial and governance information for Chinese listed firms [61]. Following the definitions by Anderson and Reeb (2003) [62], we identified family firms as those where a family member controls the company, with relatives as shareholders or directors in the listed company or its controlling entity.

We collected data from multiple sources. Initially, we extracted the personal details of founders, including their names, ages, genders, and other individual information, as well as firm-level financial details, from the Family Business Research sub-dataset within the CSMAR database. We then manually compiled data on the founders' tenures and cities of birth, based on the list of family founders, and verified this information through searches on Baidu, Sina, and the respective firms' official websites. Additionally, using the founders' birthplace data, we gathered genealogical data on their birthplaces. This information was sourced from the 'General Catalogue of Chinese Genetic Pedigrees', published by the Shanghai Library in 2008, which contains observations on approximately 38,429 clans. Last, we acquired data on the firms' green innovation patents from the State Intellectual Property Office of China (SIPO) database. These databases have been extensively utilized in prior research on Chinese listed companies [14].

We excluded observations in the following categories: (1) special treatment (ST) firms, which include financial firms and those with an ST* designation; and (2) instances where key variables had missing values. The final sample comprised an unbalanced panel of 1678 firms, amounting to 9048 firm–year observations. These firms were distributed across 22 provinces, 5 autonomous regions, and 4 municipalities on the Chinese mainland, excluding Hong Kong, Macau, and Taiwan. Table 1 presents the distribution of the sample across selected companies. A significant portion of our sample (53.28%) consisted of firms based in the east China region (i.e., Shanghai, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, and Shandong provinces), while only 2.14% of the firms were located in the northwest China region (i.e., Shanxi, Gansu, Qinghai, Ningxia, and Xinjiang provinces). The east continues to be China's economic center, contributing to more than half the nation's GDP. The average age of firms was around 17.57 years.

Table 1. Sample distribution.

A. Regional Distribution of Sample Firms	Firm Size	Firm Age	Citations of Green Patents	Freq.	%
North China Region	21.865	16.888	0.749	740	8.179
Northeastern China Region	21.781	18.056	0.557	287	3.172
East China Region	21.785	17.517	0.630	4821	53.282
Central China Region	21.786	17.640	0.844	2616	28.912
Southwest China Region	21.825	17.087	0.742	390	4.310
Northwest China Region	21.873	18.851	0.645	194	2.144
B. Year distribution of sample firm					
2012	21.375	13.380	0.374	702	7.758
2013	21.522	14.472	0.555	699	7.725
2014	21.626	15.433	0.673	743	8.212
2015	21.716	16.232	0.737	872	9.376
2016	21.835	17.127	0.772	1004	11.096
2017	21.846	17.921	0.710	1214	13.417
2018	21.916	18.888	0.810	1259	13.915
2019	21.837	19.789	0.818	1240	13.705
2020	21.986	20.236	0.692	1315	14.534

(North China region: Beijing, Tianjin, Hebei, Shanxi, and Neimenggu; northeastern region: Liaoning, Jilin, and Heilongjiang; east China region: Shanghai, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, and Shandong; central China region: Henan, Hubei, Hunan, Guangdong, Guangxi, and Hainan; southwest region: Chongqing, Sichuan, Guizhou, Yunnan, and Xizang; northwest region: Shanxi, Gansu, Qinghai, Ningxia, and Xinjiang. The regional categories are based on the resource of the China Bureau of Statistics).

3.2. Measurement

3.2.1. Dependent Variable

Firms' green innovation performance. Following prior studies on green innovation [63], we measured firms' green innovation performance using the natural logarithm of one plus the number of green patent citations.

3.2.2. Independent Variable

Founder CEOs' clan cultural values. To measure the clan cultural values of founder CEOs, we used the number of genealogies compiled in the founders' birthplaces (at the city–regional level) per million people, expressed in logarithmic form. Extensive prior studies have shown that genealogies are an effective measure of the strength of clan culture. Genealogies, as classic documents recording the origins and deeds of clans, have become a core medium for transmitting clan culture [37]. In regions with a strong clan culture, individuals tend to maintain and frequently update their genealogies, demonstrating their commitment to preserving their clan's history and traditions. Therefore, clan members in regions with a strong clan culture tend to maintain more genealogies and frequently engage in 'revising genealogy' [64]. The higher this ratio is, the more pronounced the clan cultural values are in a founder CEO's background.

3.2.3. Moderating Variables

Family involvement. Following previous research on family businesses [65], we measured family involvement by examining the proportion of family members within the top management team.

Government environmental regulation. To measure government environmental regulation, we utilized information from the government work reports of each province. These reports typically review past accomplishments and outline future directions, aiming to accurately convey the government's priorities and intentions [66]. Therefore, we posit that a higher frequency of environmental keywords mentioned in a government work report may indicate a stronger emphasis on environmental issues. We calculated government

environmental regulation by dividing the number of environmental word frequencies in the annual government work report by the total number of words. This metric allowed us to assess the regulatory environment in which a company operates and its potential impact on the company's environmental performance.

3.2.4. Control Variables

Following prior research, we accounted for variables at the founder CEO, firm, and regional levels that could impact a firm's green innovation. At the CEO level, we controlled for age (measured in years), gender (1 = male; 0 = female), and educational level (six-point scale from technical secondary school to doctorate). In addition, prior studies suggest that top managers' social networks (e.g., political ties) can provide more effective monitoring and thus affect founder's decisions [67]. Thus, we controlled for political ties, a binary variable indicating whether founders served as a representative for the Chinese People's Political Consultative Conference or the People's Congress (1 = yes; 0 = no).

At the firm level, we considered attributes affecting green innovation, such as firm size (log of total assets), age (log of years since establishment), financial performance (Tobin's Q for market value) [68], and board independence. We also controlled for the largest shareholder of the company to control for the degree of the diversification of the company. To further account for the effects of organizational slack on a firm's innovation capabilities [69], we included the firm's cash holdings as a variable in the regression model. Cash holdings, as a key measure of slack, provide firms with the flexibility to invest in new projects, explore innovative ideas, and absorb risks associated with innovation [70]. Including cash holdings in our model allows us to control the effects of financial slack on firms' innovation capacity, particularly in dynamic and resource-intensive contexts such as green innovation [69].

Furthermore, we controlled for industrial-level factors affecting green innovation and incorporated the Herfindahl–Hirschman Index (HHI) as a key variable in our research model. The HHI is a commonly used indicator of market concentration that measures the level of competition within an industry. It is calculated by summing the squares of the market shares of all firms in the market [71]. A higher HHI reflects lower competition, while a lower HHI indicates a more competitive industry structure. Including the HHI in our model was essential for understanding how market concentration influences firms' incentives and their ability to invest in green innovation [72,73].

Finally, at the regional level, considering that innovation is heavily affected by regional-level factors, we also controlled for regional economic development, measured by the natural logarithm of the regional GDP per capita, and the provincial market environment, following previous studies, e.g., ref. [74], to use the marketization index compiled by Fan et al. (2001) [75] to measure China's regional formal institutional development. To account for the potential influence of other cultural ideologies on CEOs' strategic decisions regarding green innovation, we also controlled for the effects of Confucianism, which we measured by the number of Confucian temples in the regions where the companies are registered.

3.3. Statistical Approach

We analyzed our panel data with Stata 14.1 using a fixed-effects regression model to manage variables that varied across entities but not over time and vice versa. This approach effectively minimized potential endogeneity issues with explanatory variables [76]. Additionally, to account for temporal and industrial variations, we incorporated eight year dummies and seventeen industry dummies based on the SIC code in our regression models. This method ensures that the estimated average of the dependent variable could differ across groups, effectively managing unobserved heterogeneity [77].

4. Results

4.1. Hypothesis Testing

Table 2 presents the descriptive statistics and the Pearson correlations of the variables used in the regression models. For all variables, the absolute value of the correlation coefficient does not exceed 0.4, which means that the magnitude of the correlations is low. Variance inflation factors (VIFs) were also inspected in our models. The highest VIF value is 4.88, which is well below the threshold of 5, indicating that there is no severe multicollinearity among our variables [78]. As expected, a founder's clan cultural values has a positive relationship with green innovation performance (r = 0.041; $p < 0.001$).

Table 3 presents the results of our fixed-effects hierarchical regression. Model 1 is the baseline model that includes only control variables. Model 2 adds the independent variable (founders' clan cultural values), and Model 3 incorporates the moderating variables. Model 4 is the full model, which includes two interaction terms. Hypothesis 1 posits that a founder CEO's adherence to clan cultural values positively influences a firm's green innovation practices. Model 2 in Table 3 supports the hypothesis, indicating that a founder CEO's clan cultural values exert a positive and statistically significant influence on green innovation ($\beta = 0.012$; $p < 0.01$). After introducing moderators and interaction terms, the positive effect of CEOs' clan cultural values remains robust ($\beta = 0.012$; $p < 0.01$), as demonstrated in Model 4. Therefore, Hypothesis 1 is supported.

Hypothesis 2 predicts a significant moderating effect of family involvement on the relationship between a founder CEO's clan cultural values and a firm's green innovation performance. In Model 4 of Table 3, the coefficient for the interaction term between the founder's clan cultural values and family involvement is negative and significant ($\beta = -0.093$; $p < 0.001$). To further understand the moderating effects of family involvement, we plot a two-way interaction in Figure 1. The results reveal that, in general, a firm's green innovation performance is better when family involvement is low compared to when it is high. More notably, the positive effect of CEOs' clan cultural values on firms' green innovation turns negative when family involvement is high. One possible explanation for this finding is that a higher level of family involvement introduces agency costs, particularly due to conflicts among different family members within the management team. In this context, CEOs driven by clan cultural values may work harder to consolidate their family's interests, pressuring family members to sacrifice personal or sub-family interests. This, in turn, may exacerbate intra-family conflicts as different family members pursue divergent goals [79,80]. These conflicts could undermine decision-making efficiency and reduce CEOs' commitment to green innovation.

Furthermore, when family involvement is high, the need for consensus and stability may hinder a CEOs' motivation to adopt novel, high-risk green technologies [68]. The emphasis on family unity and loyalty may lead to more conservative decision-making, as CEOs with strong clan cultural values might prioritize preserving the family's legacy and cohesion over pursuing disruptive innovations that could alter their family's traditional business model [81]. Overall, the results from Figure 1 provide further support for Hypothesis 2, which proposes that family involvement diminishes the positive influence of CEOs' clan cultural values on green innovation. The interaction figure was constructed by plotting the effects of clan culture on green innovation at two levels of family involvement: high (+1 standard deviation above the mean) and low (−1 standard deviation below the mean). This method, which used ±1 SD to visualize moderation effects [82], illustrates that the relationship between clan culture and green innovation is stronger when family involvement is low but weakens significantly when family involvement is high. By anchoring the slopes at these standardized values, the figure clarifies how the moderating variable (family involvement) alters the direction and magnitude of the focal relationship.

Table 2. Descriptive statistics and Pearson correlation.

Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Founder's gender	0.945	0.229																
2. Founder's age	4.007	0.159	0.087 ***															
3. Founder's education	3.152	1.665	0.003	−0.260 ***														
4. Political tie	0.454	0.498	0.040 ***	0.128 ***	0.018 *													
5. Firm size	21.795	1.044	−0.001	0.108 ***	−0.017	0.153 ***												
6. Firm age	2.811	0.344	0.030 ***	0.218 ***	−0.045 ***	0.009	0.155 ***											
7. Firm Tobin's Q	2.583	2.875	0.004	−0.101 ***	0.039 ***	−0.084 ***	−0.387 ***	−0.050 ***										
8. Firm TOP1	33.614	13.607	−0.007	−0.063 ***	−0.073 ***	−0.026 **	0.000	−0.077 ***	0.030 ***									
9. Firm cash-holding	19.678	1.193	−0.008	0.055 ***	0.046 ***	0.044 ***	0.712 ***	0.060 ***	−0.203 ***	0.066 ***								
10. HHI	0.136	0.142	−0.013	−0.015	−0.052 ***	−0.010	0.062 ***	0.009	0.011	0.030 ***	0.023**							
11. Capital GDP	11.250	0.381	−0.030 ***	0.072 ***	−0.020 *	−0.151 ***	0.052 ***	0.167 ***	−0.007	−0.003	0.038 ***	0.051 ***						
12. Market environment	8.728	1.422	0.001	0.016	−0.030 ***	−0.098 ***	0.010	0.039 ***	0.032 ***	0.032 ***	0.027 **	0.031 ***	0.776 ***					
13. Confucian temples	70.495	33.030	0.047 ***	0.099 ***	−0.046 ***	0.146 ***	0.025 **	−0.010	−0.041 ***	−0.035 ***	−0.014	−0.041 ***	−0.440 ***	−0.112 ***				
14. Clan cultural values	8.001	2.354	−0.005	0.039 ***	−0.139 ***	0.029 ***	0.040 ***	−0.007	−0.042 ***	0.062 ***	0.048 ***	0.027 ***	0.149 ***	0.276 ***	0.123 ***			
15. Family involvement	0.180	0.162	0.002	0.004	−0.031 ***	−0.032 ***	−0.217 ***	−0.064 ***	0.053 ***	0.067 ***	−0.117 ***	−0.009	0.048 ***	0.105 ***	0.014	0.057 ***		
16. Government regulation	0.284	0.114	0.001	−0.165 ***	0.059 ***	0.024 **	−0.086 ***	−0.232 ***	0.066 ***	0.044 ***	−0.022 **	−0.006	−0.333 ***	−0.161 ***	−0.042 ***	−0.149 ***	−0.008	
17. Green innovation	0.705	1.146	0.004	0.078 ***	0.045 ***	0.064 ***	0.392 ***	0.041 ***	−0.141 ***	−0.107 ***	0.278 ***	−0.040 ***	0.046 ***	0.053 ***	0.025 **	0.041 ***	−0.167 ***	−0.018 *

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 3. Fixed-effects hierarchical regression result.

Variables	Model (1)	Model (2)	Model (3)	Model (4)
Founder's gender	−0.060 (0.048)	−0.060 (0.048)	−0.063 (0.048)	−0.063 (0.048)
Founder's age	0.181 * (0.072)	0.184 * (0.072)	0.193 ** (0.072)	0.190 ** (0.072)
Founder's education	0.037 *** (0.006)	0.039 *** (0.006)	0.038 *** (0.006)	0.037 *** (0.006)
Founder's political tie	−0.017 (0.023)	−0.019 (0.023)	−0.019 (0.023)	−0.017 (0.022)
Firm size	0.496 *** (0.019)	0.497 *** (0.019)	0.467 *** (0.019)	0.465 *** (0.019)
Firm age	−0.022 (0.034)	−0.020 (0.034)	−0.031 (0.034)	−0.026 (0.034)
Firm Tobin's Q	0.006 (0.004)	0.007 (0.004)	0.005 (0.004)	0.005 (0.004)
Firm Top1	−0.009 *** (0.001)	−0.009 *** (0.001)	−0.008 *** (0.001)	−0.008 *** (0.001)
Firm cash-holding	−0.004 (0.012)	−0.006 (0.012)	0.002 (0.012)	0.003 (0.012)
HHI	−0.348 *** (0.106)	−0.355 *** (0.106)	−0.324 ** (0.105)	−0.330 ** (0.104)
Capital GDP	−0.135 (0.075)	−0.132 (0.075)	−0.162 * (0.075)	−0.154 * (0.075)
Market environment	0.057 *** (0.016)	0.051 *** (0.016)	0.066 *** (0.016)	0.064 *** (0.016)
Confucian Temples	−0.0001 (0.0004)	−0.0002 (0.0004)	−0.0003 (0.0004)	−0.0002 (0.0004)
Founder CEO's clan culture		0.012 ** (0.005)	0.015 ** (0.005)	0.012 ** (0.005)
Family involvement			−0.688 *** (0.063)	−0.686 *** (0.063)
Government environmental regulation			0.270 * (0.120)	0.258 * (0.119)
Founder's clan culture × family involvement				−0.093 *** (0.025)
Founder's clan culture × government environmental regulation				0.077 * (0.039)
Industry fixed	YES	YES	YES	YES
Year fixed	YES	YES	YES	YES
Observations	9048	9048	9048	9048
R^2	0.244	0.245	0.254	0.255

Notes: robust standard errors are in parentheses; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Hypothesis 3 predicted that the intensity of government environmental regulation would strengthen the positive influence of a founder CEO's clan cultural values. In Model 4 of Table 3, the results show that the coefficient for the interaction between a founder's clan cultural values and government environmental regulation is positive and significant

(β = 0.077; $p < 0.05$). Therefore, Hypothesis 3 is supported. Figure 2 illustrates the moderating effect of government environmental regulation. The interaction was plotted using Dawson's (2014) [82] mean split method, where values of the moderator (government environmental regulation) were categorized as high (+1 standard deviation above the mean) and low (−1 standard deviation below the mean). This approach clarifies how the relationship between a founder CEO's clan culture and green innovation varies across distinct regulatory conditions. Specifically, the slope for clan culture is steeper under stringent environmental regulation (high condition), whereas the effect weakens when regulation is lenient (low condition). By anchoring the moderation effects at these standardized values, the figure demonstrates that stringent regulation amplifies the influence of clan culture, thereby supporting Hypothesis 3.

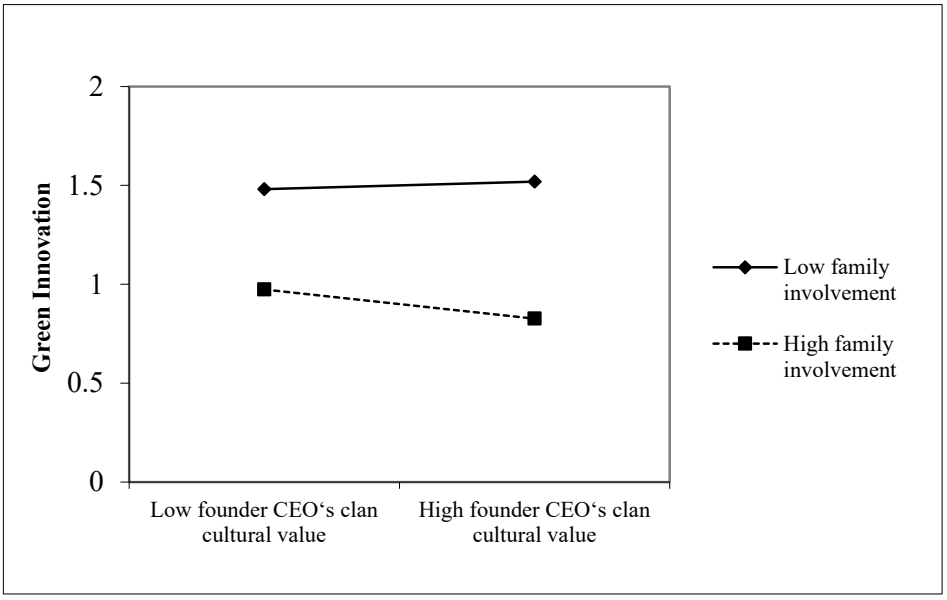

Figure 1. The moderating effect of family involvement.

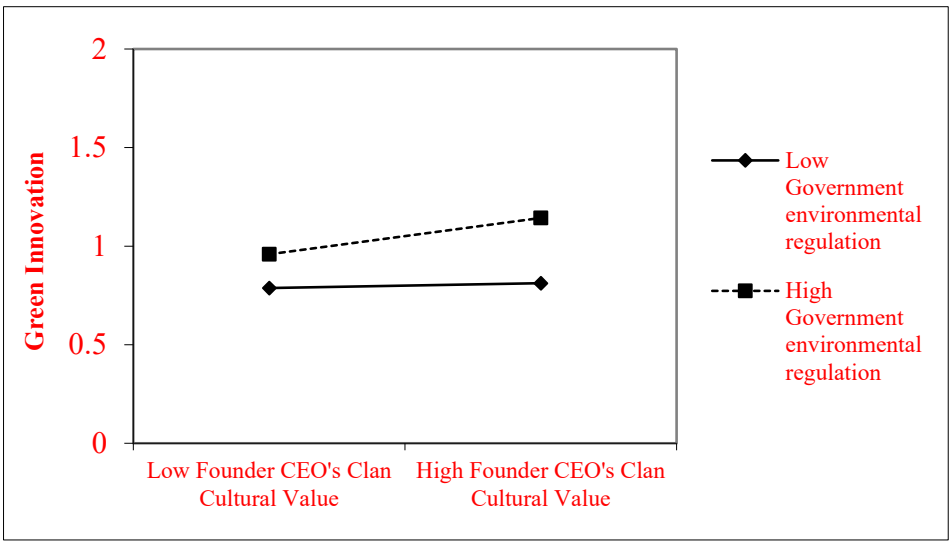

Figure 2. The moderating effect of government environmental regulation.

4.2. Robustness Test

To corroborate our findings and gauge the robustness of our results, we conducted several supplemental tests. First, we employed second-generation involvement as an alternative measure of family involvement. Previous research shows that second-generation

involvement often leads to power shifts and cultural changes, affecting a firm's innovation goals [83]. We created a binary variable, assigning '1' if a founder CEO's descendant works in the family business and '0' otherwise, to measure second-generation involvement as an alternative indicator of family involvement [84]. Our findings remained consistent when employing this alternative measure of family involvement. Moreover, we conducted regressions on a randomly chosen subsample that constituted 85% of all observations. The outcomes were completely aligned and robust, matching those derived from the entire sample. These results further confirmed that our previous findings were robust.

5. Discussion

This study provides a robust theoretical framework for examining the complex dynamics that influence green innovation performance within family firms. Our results show that a founder CEO's clan cultural values have a significant positive effect on a firm's green innovation performance, and this relationship is weakened by family involvement and amplified by government environmental regulation.

Despite a significant body of prior research exploring the direct positive effect of family involvement on a firm's green innovation and environmentally friendly practices—due to a long-term orientation and a commitment to maintaining a responsible family identity e.g., [85–88]—our study finds a negative influence of family involvement on initiating green innovation, based on the behavioral agency perspective and socioemotional wealth (SEW). Specifically, we suggest that the positive relationship between CEOs' clan cultural values and green innovation is constrained when family involvement is high, as differentiated family goals emerge. Extended family members, particularly those not directly related by blood, may prioritize short-term financial gains, such as dividends, over long-term strategic objectives [42,43]. This divergence in goals can lead to a misalignment between the founder's long-term vision and the objectives of other family members, further exacerbating agency costs and making it difficult to implement long-term strategies such as green innovation [41,89]. We discuss the contributions of these findings in more detail in the following sections.

5.1. Theoretical Contributions

This study contributes to the management literature in several ways. First, by investigating the effects of founder CEOs' clan cultural values on firms' green innovation, our study contributes new empirical evidence demonstrating that a firm's strategic orientation toward environmental innovation is significantly shaped by the values and beliefs of its founder CEO, as outlined in upper-echelon theory. This theory suggests that top executives' personal experiences, cognitive biases, and values deeply influence the strategic decisions and outcomes of the organizations they lead [26]. However, despite its theoretical prominence, there has been limited empirical focus on the role of CEOs' cultural values, particularly in the context of family businesses [90]. Our research addresses this gap by integrating clan cultural values into the upper-echelon framework, thereby extending the traditional focus of upper-echelon theory, which has predominantly concentrated on demographic characteristics, personalities, and professional experiences [14].

Although some scholars have explored the role of CEOs' cultural values—including Confucianism and broader national cultural influences such as Hofstede's framework—in shaping a firm's environmental strategies [91,92], the specific role of clan culture, particularly in the Asian context, remains underexplored in the green innovation literature. Confucianism, for instance, emphasizes hierarchical relationships, loyalty to the state, and the importance of education [93], while Hofstede's framework focuses on broad dimensions such as individualism versus collectivism and power distance [94]. In contrast,

clan culture places greater emphasis on loyalty, long-term relationships, intergenerational responsibility, and the primacy of family ties, values that are often passed down through generations and have a lasting impact on a firm's strategy, particularly in family businesses. In family-owned businesses, especially those led by founder CEOs, clan cultural values are especially influential. These firms are often characterized by strong familial bonds, a long-term strategic orientation, and a focus on community welfare, all of which are rooted in clan culture [22]. Unlike other cultural frameworks, clan culture helps explain the heterogeneity of family business behaviors, offering critical insights into how deeply ingrained family values shape decision-making, particularly with regard to environmental responsibility. Clan cultural values not only help explain the diversity of family business behaviors in their approach to green innovation but also provide a nuanced understanding of the psychological dynamics that drive leadership in these contexts. Our findings enrich the current scholarly discourse by connecting the psychology of leadership with the pursuit of environmental sustainability within the realm of strategic management.

In addition, although the previous literature on imprinting theory suggests that founder's value imprints can decay or amplify by subsequent experiences and environment, there is a lack of empirical studies on the contingency context of founder imprints [11]. Our study expands the existing literature on founder imprints by highlighting the potential evolution of founders' clan culture imprints due to family involvement and changes in government policy [10]. Our research adds to this body of knowledge by emphasizing the importance of considering family-related contingent factors to understand the dynamic change in founder imprints in the family business context.

Moreover, our study contributes to the SEW and family business literature on agency problems by examining the interaction effects between founder CEOs' clan cultural values and family involvement. While traditional agency theory typically focuses on conflicts between owners and managers, in family firms, these conflicts can also extend to various family members, each with potentially divergent interests and levels of influence [50]. Our results indicate that two distinct types of family conflicts can arise due to high levels of family involvement in business management: conflicts of interest among sub-families and conflicts stemming from differing opinions among family members. These conflicts increase agency costs and reduce the decision-making efficiency of clan-oriented founder CEOs in the context of green innovation.

Although family firms are traditionally seen as entities that seek to preserve and enhance SEW—often focusing on long-term goals that align with the founder's vision [23,46]—high levels of family involvement may induce heterogeneous family goals. We suggest that diverse family member interests, especially among those in top management positions, ranging from wealth preservation to family control or reputation maintenance, can lead to fragmented SEW [68,95]. This fragmentation weakens the family's ability to pursue unified long-term objectives, particularly when facing challenges such as green innovation, which typically requires substantial long-term investments [96]. The presence of sub-family factions or branches within the business can lead to intra-family conflicts, as different groups may prioritize their immediate interests over the broader family or company's long-term goals [43], thereby diluting the positive effects of the founder's clan cultural values. These findings contribute to a deeper understanding of the heterogeneity of SEW and the agency costs associated with broader family and sub-family groups, an aspect that has been largely overlooked in the previous family business literature.

5.2. Practical Implications

This study provides valuable insights for practitioners and policymakers interested in promoting green innovation and achieving sustainable development goals. First, the

positive influence of CEO clan culture on green innovation carries significant implications for family businesses seeking to enhance their sustainable practices. For these firms, it is essential to cultivate and empower leaders who demonstrate clan-oriented leadership qualities. Such leaders have the ability to foster a robust organizational culture that promotes knowledge-sharing, collaboration, and a steadfast commitment to long-term sustainability goals. Prior research has shown that CEOs who prioritize the well-being of both their family and their firm cultivate a culture of mutual trust, which positively impacts firm performance, including areas such as green innovation [79,97]. To leverage this, family businesses should consider investing in leadership development programs that reinforce the values of clan culture while integrating green innovation practices. This approach will help establish sustainability as a central pillar of their corporate strategy.

Our findings also indicate that family involvement in management attenuates the positive impact of founder CEOs' clan cultural values on firms' green innovation performance. This nuanced interplay underscores the importance of understanding the specific dynamics of family governance and the individual objectives of family members in promoting a culture that supports green innovation within family firms. To mitigate the agency costs associated with high family involvement, it is crucial to implement effective governance mechanisms and strategically align family members' interests with firms' long-term goals. We recommend that family firms develop clear governance structures that delineate family interests from operational decisions. This separation ensures that innovation efforts—particularly those related to green technologies—are not unduly constrained by familial involvement. Establishing such governance frameworks can foster a more conducive environment for green innovation by reducing potential conflicts and aligning the strategic direction of firms with sustainable practices.

The positive moderating effect of government regulation suggests that policymakers play a crucial role in fostering green innovation by creating an environment that is both supportive and incentivizing. Regulations that promote sustainable practices, such as tax incentives for green technology adoption or grants for environmental R&D, can provide the necessary motivation for family firms to engage in green innovation without shouldering the financial burden typically associated with such initiatives [98]. Additionally, policymakers should consider expanding collaborative platforms that bring together government agencies, private enterprises, and academic institutions to facilitate green innovation. Such platforms can equip family firms with essential resources, including access to research partnerships and subsidized R&D programs. These public–private collaborations are particularly beneficial for family firms that may lack the internal capabilities to conduct extensive green innovation research independently [99].

5.3. Limitations and Future Directions

Our study is not without limitations, which provide avenues for future research. Firstly, our data were solely gathered from China, a prominent emerging market, which may constrain the generalizability of our findings to broader contexts. Clan culture emphasizes values such as loyalty, cooperation, shared identity, and family-like relationships within an organization [100]. This cultural orientation aligns closely with collectivist societies, such as China, where group harmony and mutual support are prioritized over individual achievement. In comparison to other countries, clan culture is often more deeply ingrained in Chinese family firms, where personal relationships, familial ties, and a long-term orientation are integral to decision-making processes [101]. In these contexts, CEOs are likely to exhibit a strong commitment to green innovation, driven by a deep sense of responsibility toward future generations, family legacy, and broader social welfare [102]. In contrast, in more individualistic cultures, such as in the United States or

Western Europe, clan culture is less embedded within the organizational framework. Here, CEOs may prioritize short-term financial goals over collective welfare [94]. While family firms are still common in these regions, their governance structures tend to be more formal, with less emphasis on the clan-like values that shape decision-making in family-owned businesses. For example, in the U.S., although some family firms, like Patagonia, are recognized pioneers in green innovation, many others may approach sustainability more pragmatically, driven by market advantages rather than a sense of familial duty or collective benefit [102]. In countries with stringent environmental regulations, CEOs are often subjected to strong institutional pressures that compel them to innovate in environmentally sustainable ways [103]. In these regions, green innovation may be seen less as a voluntary strategic choice driven by personal cultural values and more as a regulatory necessity. While China exemplifies a clear case of clan culture, other Asian countries, such as South Korea and Japan, share some cultural similarities. For instance, in Japan, where family businesses are often long-standing and rooted in traditional values, CEOs may adopt a more holistic approach to green innovation that reflects both the importance of family legacy and a sense of social responsibility [103].

To assess the broader applicability of our findings, we encourage future research to explore how the clan culture of founders influences green innovation in various cultural contexts. Comparative studies examining family firms in different countries with varying cultural orientations and regulatory environments would deepen our understanding of how CEOs' cultural values shape strategic decisions related to sustainability. In particular, we suggest conducting comparative studies between Chinese family firms and those in Western regions, such as North America and Europe, with varying degrees of government involvement. Future research could employ multi-country datasets to explore whether similar patterns emerge in different economic and regulatory environments.

Another limitation of this study lies in the measurement of core variables, particularly the proxies used to capture clan culture and government regulation, which are constrained by the use of publicly available data sources. While the use of innovative proxies, such as genealogy compilations for clan culture and keyword frequency in reports for government regulation, offers unique insights and has been widely employed in previous studies e.g., [104], these proxies may not fully capture the depth and complexity of these constructs. For instance, relying on genealogy compilations may only provide a limited representation of clan culture, as it fails to account for more dynamic aspects of organizational culture, such as interpersonal relationships, informal communication, and the day-to-day interactions that characterize a clan-like environment. Therefore, we suggest that future research focus on developing a valid, multi-item scale to measure CEOs' clan cultural values, which would more effectively capture the complexity of this construct.

Given that triangulating multiple methods enhances the validity of constructs and strengthens the robustness of findings [105], we suggest that future studies combine quantitative proxies with qualitative methods, such as surveys or interviews with top executives (e.g., CEOs), to gain a more nuanced understanding of how clan culture manifests in practice. Similarly, exploring government regulation through a broader range of qualitative data sources would offer a deeper understanding of its impact on organizations. Future research could involve expert interviews, case studies, or even ethnographic methods to capture the lived experiences of organizations navigating complex regulatory environments [106]. Utilizing a mixed-methods approach would enable researchers to triangulate data from multiple sources, thereby enhancing the validity and reliability of findings.

6. Conclusions

Leveraging upper-echelon, imprinting, and agency theories, this study highlights how founder CEOs' clan cultural values drive green innovation, weakened by family involvement and strengthened by government environmental regulation. This approach reveals the deep impact of founder CEOs' values, family dynamics, and government policies on the strategic green innovation outcomes of family firms. Our framework emphasizes the importance of viewing founder CEOs' values, family engagement, and government policies as interlinked factors shaping the strategies and results of family firms.

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